

Stat Phys 2 21/01

$N \gg 1$, thermo limit equilibrium: $\left\{ \begin{array}{l} \text{microcanonical} \\ \text{canonical} \\ \text{grand canonical} \end{array} \right.$ ensembles $\begin{array}{l} E = \text{const} \quad N = \text{const} \\ \langle E \rangle \approx E_F \approx \text{const} \end{array}$

Systems: (1) Harmonic Oscillators
- sound wave propagation in a solid

(2) Free system, particles in a "box"
Fermions & bosons

(3) Magnetic Dipoles, localised or not no momentum

(4) Ising Model in 1 dimension

Microcanonical	Canonical	Grand Canonical
Totally isolated E, V, N conserved	Exchange heat $T \approx \text{const}$ $T, V, N \sim \text{conserved}$ $Z(T, V, N)$ $F = -kT \ln(Z)$ $P = -\frac{\partial F}{\partial V} \quad S = -\frac{\partial F}{\partial T} \quad \mu = \frac{\partial F}{\partial N}$	Exchange heat / particles $T, \mu \approx \text{const}$ $(T, V, \mu) \sim \text{conserved}$ $\mathcal{Z}(T, V, \mu)$ $J = -kT \ln(\mathcal{Z})$ $S = -\frac{\partial J}{\partial T} \quad P = -\frac{\partial J}{\partial V}$ $N = -\frac{\partial J}{\partial \mu}$

In thermo limit the physics at eqbm are $\approx$ the same in all ensembles

$\Rightarrow$ (M) $E = \text{const}$

(G-C) $\langle E \rangle \approx \text{const} = E_F$ most probable value of E

(C) $\langle E \rangle \approx \text{const} = E$

$\langle N \rangle \approx \text{const} \approx N_F$

We will be dealing with quantum systems with discrete energy spectrums

Heisenberg's uncertainty principle $\hbar \neq 0$ (q, p) not appropriate variables since there is no precise values for the trajectory

Microcanonical

$P_n = \frac{1}{\Omega(E, V, N)}$, where n denotes the set of quantum numbers that completely determine a microstate

Example Harmonic Oscillator

A system which contains N one-dimensional h.o with the same frequency

$$\epsilon_i = (n_i + \frac{1}{2}) \hbar \omega$$

$$E = \sum_{i=1}^N \hbar \omega (n_i + \frac{1}{2}) = \frac{1}{2} N \hbar \omega + \hbar \omega \sum_{i=1}^N n_i$$

Ω is effectively the number of all possible combinations of $\{n_i\}$ s.t.

$$\sum_{i=1}^N n_i = (\frac{E}{\hbar \omega} - \frac{N}{2}) \equiv \tilde{E}$$

$$\Rightarrow \Omega(E, V, N) = \sum_{\{n_1, \dots, n_N\} \in \mathcal{N}} \delta(\sum_{i=1}^N n_i, \tilde{E})$$

$$W(E, V, N) \Delta E = \int_{\substack{d\mathbf{q} d\mathbf{p} \\ E < H < E + \Delta E}} \frac{d\mathbf{q} d\mathbf{p}}{h^3} \quad (\text{classical})$$

Canonical

In this case, the probability of a given microstate is

$$p_n = \frac{e^{-\beta E_n}}{Z(T, V, N)}, \quad \beta = \frac{1}{kT}$$

$$\sum_n p_n = 1 = \frac{1}{Z} \sum_n e^{-\beta E_n} \Rightarrow Z(T, V, N) = \sum_n e^{-\beta E_n}$$

$$\downarrow \text{(classical)} \quad Z = \int \frac{d\mathbf{q} d\mathbf{p}}{h^3} e^{-\beta H}$$

Here is any function of E , not F should have written $\mathcal{F}$

Recall that we showed:

$$\int_{\text{all phase space}} \frac{d\mathbf{q} d\mathbf{p}}{h^3} e^{-\beta H(\mathbf{q}, \mathbf{p})} \mathcal{F}(\mathbf{q}, \mathbf{p}) = \langle \mathcal{F} \rangle = \int_{\text{all energies}} dE W(E) e^{-\beta E} \mathcal{F}(E)$$

$$\text{intuitively} \Rightarrow \sum_n e^{-\beta E_n} \mathcal{F}(E_n) = \sum_{E=0}^{\infty} \mathcal{F}(E) e^{-\beta E} \Omega(E) = \langle \mathcal{F} \rangle$$

Can we write the above in bra-ket notation?

$$Z = \sum_n e^{-\beta E_n} = \sum_n \langle n | e^{-\beta \hat{H}} | n \rangle \equiv \text{Tr} e^{-\beta \hat{H}} \quad |n\rangle : \text{eigenstates of } \hat{H}$$

$$\hat{\rho} \equiv e^{-\beta \hat{H}} : \text{thermal density operator}$$

$$\langle X \rangle = \sum_n \langle n | e^{-\beta \hat{H}} \hat{X} | n \rangle$$

Grand Canonical

$$P_{\Omega, N} = \frac{1}{\mathcal{Z}} e^{\beta \mu N} Z(T, V, N)$$

replace with discrete sum

$$J = -KT \ln \mathcal{Z}$$

$$\mathcal{Z} = \sum_{N=0}^{\infty} e^{\beta \mu N} Z(T, V, N)$$

Example Harmonic Oscillators

study the system of $1d$ h.o in the thermodynamic limit.

This implies that we can use our preferred ensemble.

microcanonical usually more difficult, use $\mathcal{C}$ or $\mathcal{G}$ first.

h.o.s are not identical here (but assume ω the same for all)

We will start by considering the partition function.

$$Z(T, V, N) = \sum e^{-\beta E_0}$$

$$E_0 = \sum_{i=1}^N (n_i + \frac{1}{2}) \hbar \omega = \sum_{i=1}^N \epsilon(n_i) \rightarrow Z = \sum_{(n_1, \dots, n_N)} e^{-\beta \sum_{i=1}^N \epsilon(n_i)}$$

$$= \sum_{\{n_1, \dots, n_N\}} e^{-\beta \epsilon(n_1)} \dots e^{-\beta \epsilon(n_N)}$$

$$= \sum_{n_1} e^{-\beta \epsilon(n_1)} \dots \sum_{n_N} e^{-\beta \epsilon(n_N)}$$

$$= \prod_{i=1}^N \left(\sum_{n_i} e^{-\beta \epsilon(n_i)} \right) = \prod_{i=1}^N J_i$$

$$\text{but } J_i = J = \sum e^{-\beta \epsilon(n)}$$

$$\Rightarrow Z = J^N$$

Now we must compute J for each h.o

$$J = \sum_n e^{-\beta \epsilon(n)} = \sum_n e^{-\beta \hbar \omega (n + \frac{1}{2})} = \sum_{n=0}^{\infty} e^{-\beta \hbar \omega (n + \frac{1}{2})} = e^{-\beta \hbar \omega / 2} \sum_{n=0}^{\infty} (e^{-\beta \hbar \omega})^n$$

geometric series ratio $e^{-\beta \hbar \omega}$

$$J = \frac{e^{-\beta \hbar \omega / 2}}{1 - e^{-\beta \hbar \omega}} \rightarrow Z = \left(\frac{e^{-\beta \hbar \omega / 2}}{1 - e^{-\beta \hbar \omega}} \right)^N$$

$$= \frac{1}{(e^{\frac{\beta \hbar \omega}{2}} (1 - e^{-\frac{\beta \hbar \omega}{2}}))^N} = \frac{1}{(e^{\frac{\beta \hbar \omega}{2}} - e^{-\frac{\beta \hbar \omega}{2}})^N} = \frac{1}{(2 \sinh(\frac{\beta \hbar \omega}{2}))^N}$$

Note if h.o.s had different ω s $Z = \prod_{i=1}^N \frac{1}{2 \sinh(\frac{\beta \hbar \omega_i}{2})}$

Check against classical limit: $\beta \hbar \omega \ll 1$
 $\hbar \omega \ll kT$
 $\hbar$ smallest } classical limit

$$Z = \frac{1}{2^N \sin^2(\frac{\beta \hbar \omega}{2})} \xrightarrow{\text{classical limit}} \frac{1}{2^N (\frac{\beta \hbar \omega}{2})^N} = (\beta \hbar \omega)^{-N}$$

$$F = -kT \ln Z = \frac{N \hbar \omega}{2} + kT N \ln [1 - e^{-\beta \hbar \omega}]$$

$$\Rightarrow S = kN \left\{ \frac{\beta \hbar \omega}{e^{\beta \hbar \omega} - 1} - \ln(1 - e^{-\beta \hbar \omega}) \right\} \xrightarrow{\text{check classical limit}} kN \{1 - \ln(\beta \hbar \omega)\}$$

$$P = 0 \quad M = \frac{F}{N} \quad C_V = -T \left(\frac{\partial^2 F}{\partial T^2} \right)$$

$$\langle E \rangle = F + TS = N \left\{ \frac{\hbar \omega}{2} + \frac{\hbar \omega}{e^{\beta \hbar \omega} - 1} \right\} \xrightarrow{\text{check classical limit}} NkT$$

Note: Quantum oscillators do not satisfy the equipartition of energy theorem only true in classical limit

Now consider the Microcanonical Ensemble

$$\Omega(E, V, N) = \sum_{\{n_1, \dots, n_N\}} \delta(\sum_i n_i \tilde{E} - \tilde{E}) \quad \tilde{E} = \left(\frac{E}{\hbar \omega} - \frac{N}{2} \right), \text{ some positive number}$$

We need to find how many ways we can distribute $\tilde{E}$ into N groups $\{n_1, \dots, n_N\}$

Think of $\tilde{E}$ as the number of identical balls & the N groups as the boxes we need to distribute the into (we can have empty boxes "stars & bars" think of where to place the $(N-1)$ walls in $(\tilde{E} + N - 1)$ positions)

$$\rightarrow \binom{\tilde{E} + N - 1}{N - 1} \Rightarrow \Omega = \frac{(\tilde{E} + N - 1)!}{(N - 1)! \tilde{E}!}$$

Another method to compute Ω , if you know Z

$$Z = \sum_n e^{-\beta E(n)} = e^{-\frac{\beta \hbar \omega N}{2}} \sum_{\{n_1, \dots, n_N\}} e^{-\beta \hbar \omega (n_1 + \dots + n_N)}$$

$$= e^{-\frac{\beta \hbar \omega N}{2}} \sum_{\tilde{E}=0}^{\infty} e^{-\beta \hbar \omega \tilde{E}} \sum_{\{n_1, \dots, n_N\}} \delta(\sum_i n_i \tilde{E} - \tilde{E}) = e^{-\frac{\beta \hbar \omega N}{2}} \sum_{\tilde{E}=0}^{\infty} e^{-\beta \hbar \omega \tilde{E}} \Omega(\tilde{E}, V, N)$$

$$\text{ux result for } Z \rightarrow \frac{e^{-\frac{\beta \hbar \omega N}{2}}}{(1 - e^{-\beta \hbar \omega})^N} = e^{-\frac{\beta \hbar \omega N}{2}} \sum_{\tilde{E}=0}^{\infty} e^{-\beta \hbar \omega \tilde{E}} \Omega(\tilde{E})$$

$$\sum_{\tilde{E}=0}^{\infty} \frac{1}{(1 - e^{-\beta \hbar \omega})^N} = \sum_{\tilde{E}=0}^{\infty} x^{\tilde{E}} \Omega(\tilde{E}) \xrightarrow{\text{Exercise}} \left(\frac{\tilde{E} + N - 1}{N - 1} \right)$$

StatPhys 2 Sound Wave Propagation in a Solid 30/01

- Two ways to describe:
- 1) Consider as a collection of H.O.s with given frequencies
 - 2) Consider as a gas, collection of free particles $\equiv$ "phonons"

$$1) H = V_0 + \sum_{i=1}^{3N} \frac{1}{2} (m \omega_i^2 q_i^2 + \frac{1}{m} p_i^2)$$

(rest energy)

$$QM \rightarrow E_{tot} = V_0 + \sum_{i=1}^{3N} (n_i + \frac{1}{2}) \hbar \omega_i$$

interested in the thermo-limit
so we use canonical ensemble

$$Z = e^{-\beta V_0} \prod_{i=1}^{3N} \mathcal{Z}_i, \quad \mathcal{Z}_i = \frac{e^{-\frac{1}{2} \beta \hbar \omega_i}}{1 - e^{-\beta \hbar \omega_i}}$$

$$\rightarrow Z = e^{-\beta V_0} \prod_{i=1}^{3N} \frac{e^{-\frac{1}{2} \beta \hbar \omega_i}}{1 - e^{-\beta \hbar \omega_i}} \quad \langle E \rangle = -\frac{\partial}{\partial \beta} \ln Z$$

"ground state"

$$= -\frac{\partial}{\partial \beta} (-\beta V_0 + \sum_{i=1}^{3N} \ln \mathcal{Z}_i)$$

$$= -\frac{\partial}{\partial \beta} (-\beta V_0 - \frac{\beta}{2} \sum_{i=1}^{3N} \hbar \omega_i - \sum_{i=1}^{3N} \ln(1 - e^{-\beta \hbar \omega_i}))$$

$$\beta \hbar \omega_i = \frac{\hbar \omega_i}{kT} \quad \ll 1 \Rightarrow \text{classical}$$

$\gg 1 \Rightarrow$ Quantum

$$= V_0 - \frac{\hbar}{2} \sum_{i=1}^{3N} \omega_i - \sum_{i=1}^{3N} \frac{\hbar \omega_i e^{-\beta \hbar \omega_i}}{1 - e^{-\beta \hbar \omega_i}}$$

independent of β
 $\Rightarrow$ "ground state"

$$\langle E \rangle \approx V_0 + \sum_{i=1}^{3N} \frac{\hbar \omega_i}{2}$$

$$C_V = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_{V,N} = K \sum (\beta \hbar \omega_i)^2 \frac{e^{-\beta \hbar \omega_i}}{(e^{-\beta \hbar \omega_i} - 1)^2}$$

Assume $\omega_i \approx \omega_E$

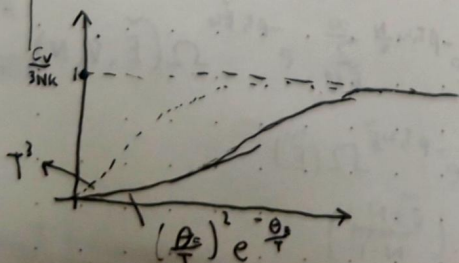
$$\rightarrow \langle E \rangle = 3N \hbar \omega_E \left(\frac{1}{2} + \frac{e^{-\beta \hbar \omega_E}}{1 - e^{-\beta \hbar \omega_E}} \right) + V_0$$

$$C_V = 3Nk \frac{x^2 e^x}{(e^x - 1)^2} = 3Nk \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\frac{\theta_E}{T}}}{(e^{\frac{\theta_E}{T}} - 1)^2}$$

$x = \beta \hbar \omega_E$ $x = \frac{\theta_E}{T}$

$$\text{let } x \gg 1 \quad C_V \approx 3Nk \left(\frac{\theta_E}{T} \right)^2 e^{-\theta_E/T}$$

$kT \ll \hbar \omega_E$
 $T \ll \theta_E$



$$C_V \sim T^3, \quad \langle E \rangle \sim T^4, \quad \ln Z \sim -\frac{1}{\beta^3}$$

$$\text{we have } \ln Z = -\left(\beta \hbar \sum_{i=1}^{3N} \omega_i + \beta P V_0\right) - \sum_{i=1}^{3N} \ln(1 - e^{-\beta \hbar \omega_i})$$

Vibrational Modes in a Solid

$$C_V \sim T^3 \text{ for all materials}$$

$$\ln Z = -\beta \hbar \sum_{i=1}^{3N} (\omega_i + V_0) - \sum_{i=1}^{3N} \ln(1 - e^{-\beta \hbar \omega_i})$$

ω_0 scale,
 $\omega_i > \omega_0$ don't contribute to C_V

$$g(\omega) = \sum_{i=1}^{3N} \int_0^\infty \delta(\omega - \omega_i) d\omega = \sum_{i=1}^{3N} 1 = 3N$$

$$\text{Approximate } \tilde{g}(\omega) : \int_0^{\omega_0} \tilde{g}(\omega) d\omega = 3N \Rightarrow g(\omega) \approx 0 \text{ for } \omega > \omega_0, \text{ assume continuous}$$

a good estimate for $\omega_0 \sim \frac{1}{a}$, $a \equiv$ interatomic distance

Waves with $\lambda \gg a$ would not propagate through the material
can think of $g(\omega)$ as an almost continuous spectrum of frequencies

$$\ln Z = -\beta \hbar \sum_{i=1}^{3N} (\omega_i + V_0) - \int_0^{\omega_0} d\omega g(\omega) \ln(1 - e^{-\beta \hbar \omega})$$

Estimate $\tilde{g}(\omega)$, Investigate homo properties of $\tilde{g}(\omega)$

$$\text{Volume fixed } \simeq N a^3 \stackrel{a \rightarrow \frac{1}{\omega_0}}{=} \frac{N L^3}{N} \left(\frac{a}{L}\right)^3, \quad \omega_0 \sim \frac{1}{a} \rightarrow \omega'_0 \simeq \omega_0$$

$$1) \int_0^{\omega_0} \tilde{g}(\omega) d\omega = 3N$$

$$2) \text{ Rescale: } \int_0^{\omega_0} \tilde{g}(\omega) d\omega = 3N L^3 \Leftrightarrow \int_0^{\omega'_0} d\tilde{\omega} \tilde{g}(\tilde{\omega}) = 3N L^3$$

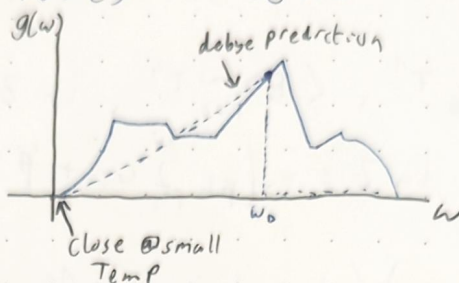
$$\int_0^{\omega_0} \tilde{g}(\omega) d\omega = 3N L^2 \Leftrightarrow \int_0^{\omega'_0} d\omega \tilde{g}(\omega) = 3N L^2$$

$$\Leftrightarrow \int_0^{\omega'_0} \tilde{g}(\omega) d\omega = \int_0^{\omega'_0} L^2 \tilde{g}(\omega) d\omega$$

$$\Rightarrow \tilde{g}(L\omega) = L^2 \tilde{g}(\omega) \Rightarrow \omega \text{ homo degree } 2$$

$$\Rightarrow \tilde{g}(\omega) = A \omega^2 \quad (\text{Euler})$$

$$\tilde{g}(\omega) = \begin{cases} 0, & \omega > \omega_D \\ \frac{a}{\omega_D} \left(\frac{\omega}{\omega_D}\right)^2, & \omega \leq \omega_D \end{cases}$$



$$\langle E \rangle = -\frac{\partial}{\partial \beta} \ln Z$$

$$= V_0 + \frac{\hbar}{2} \int_0^\infty d\omega g(\omega) \omega + \int_0^\infty d\omega \frac{g(\omega) \hbar \omega}{e^{\beta \hbar \omega} - 1}$$

$$\ln Z = -\beta V_0 - \frac{\beta \hbar}{2} \int_0^\infty d\omega g(\omega) \omega - \int_0^\infty d\omega g(\omega) \ln(1 - e^{-\beta \hbar \omega})$$

$$C_V = \frac{\partial \langle E \rangle}{\partial T} = k \int_0^\infty d\omega g(\omega) \frac{(\beta \hbar \omega)^2}{(e^{\beta \hbar \omega} - 1)^2} e^{\beta \hbar \omega} = k \int_0^{\omega_D} d\omega \left(\frac{9N}{\omega_D^3}\right) \omega^2 \frac{(\beta \hbar)^2 \omega^2}{(e^{\beta \hbar \omega} - 1)^2} e^{\beta \hbar \omega}$$

$$= k \left(\frac{9N}{\omega_D^3}\right) (\beta \hbar)^2 \int_0^{\omega_D} d\omega \frac{\omega^4 e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2} = \left(\frac{9N}{\omega_D}\right) \frac{\hbar^2}{k T^2} \int_0^{\omega_D} d\omega \frac{\omega^2 e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2}$$

$$= \frac{\hbar^2}{k T^2} \frac{9N}{\omega_D^3} \left(\frac{k T}{\hbar}\right)^5 \int_0^{\theta_D} dx \frac{x^4 e^x}{(e^x - 1)^2} \quad \theta_D = \frac{\hbar \omega_D}{k T}$$

$$C_V = 9 N k \left(\frac{T}{\theta_D}\right)^3 \int_0^{\theta_D} dx \frac{x^4 e^x}{(e^x - 1)^2}$$

for $\beta \hbar \omega \gg 1$ (small T):

$$C_V = 9 N k \left(\frac{T}{\theta_D}\right)^3 \int_0^{\theta_D} dx \frac{x^4 e^x}{(e^x - 1)^2} \quad \leftarrow \text{some number as long as it converges}$$

$\Rightarrow C_V \sim T^3$ as expected

for large $T \gg \theta_D$:

$$C_V = 9 N k \left(\frac{T}{\theta_D}\right)^3 \int_0^{\theta_D} dx \frac{x^4}{(1 - x)^2} = 9 N k \left(\frac{T}{\theta_D}\right)^3 \left(\frac{\theta_D}{T}\right)^3 \cdot \frac{1}{3}$$

$= 3 N k \rightarrow$ classical result

Stat Phys 2

Vibrational Modes in Solid 01/02

Computation of $I = \int_0^\infty dx \frac{x^4 e^x}{(e^x - 1)^2} = -\frac{x^4}{e^x - 1} \Big|_{x=0}^\infty - 4 \int_0^\infty dx \frac{x^3}{e^x - 1}$

$\Rightarrow \int_0^\infty dx \frac{x^3}{e^x - 1} = \sum_{n=0}^\infty \int_0^\infty dx x^3 e^{-(n+1)x} \stackrel{\text{I.O.P.}}{=} \sum_{n=0}^\infty \frac{6}{(n+1)^4} = \frac{\pi^4}{15}$

used: $\frac{1}{e^x - 1} = \frac{e^{-x}}{1 - e^{-x}} = e^{-x} \sum_{n=0}^\infty e^{-(n+1)x}$

Stat Phys 2 Theory of Ideal Systems of Identical Particles

13/02

$$E = \epsilon_{i_1} + \epsilon_{i_2} + \dots + \epsilon_{i_N} \text{ No longer makes sense}$$

$$E = \sum_i n_i \epsilon_i \quad \epsilon_i: \text{distinct energy level}$$

$$N = \sum_i n_i$$

"occupation number":
how many particles
occupy each level

Suppose we want to work in canonical ensemble

T or β, V, N

$$Z = \sum_{\{n_i, \dots, n_{\infty}\}} e^{-\beta \sum_i n_i \epsilon_i} \delta_{N, \sum n_i} = \begin{cases} 1, & N = \sum n_i \\ 0, & \text{otherwise} \end{cases}$$

So use Grand-canonical ensemble

$$\mathcal{Z}(T, V, \mu) = \sum_{N=0}^{\infty} e^{\beta \mu N} Z(T, V, N) = \sum_{N=0}^{\infty} e^{\beta \mu N} \sum_{\{n_i, \dots\}} e^{-\beta \sum_i n_i \epsilon_i} \delta_{N, \sum n_i} =$$

$$= \sum_{\{n_i, \dots\}} e^{-\beta \sum_i n_i \epsilon_i} \sum_{N=0}^{\infty} e^{\beta \mu N} \delta_{N, \sum n_i} = \sum_{\{n_i, \dots\}} e^{-\beta \sum_i n_i (\epsilon_i - \mu)}$$

$$\mathcal{Z} = \sum e^{-\beta \sum_i n_i (\epsilon_i - \mu)} = \sum_{n_1} \sum_{n_2} \dots e^{-\beta n_1 (\epsilon_1 - \mu)} e^{-\beta n_2 (\epsilon_2 - \mu)} \dots$$

$$\Rightarrow \mathcal{Z} = \prod_i \left(\sum_{n_i} e^{-\beta n_i (\epsilon_i - \mu)} \right) \stackrel{\text{ideal}}{\Rightarrow} \mathcal{Z} = \prod_i \mathcal{Z}_i$$

$\mathcal{Z}_i$: grand-canonical ensemble of one level i

Two Cases: ① A single energy level can be populated by any number of particles
 $n_i = 0, 1, \dots$
bosons, particles of zero or integer spin

② A single energy level can be populated by at most one particle
Fermi: Dirac statistics
Fermions, half integer spin

13/02

Ideal systems of identical particles Stat Phys

$$\delta_i = \begin{cases} \textcircled{1} \sum_{n_i=0}^{\infty} e^{-\beta n_i (\epsilon_i - \mu)} = \frac{1}{1 - e^{-\beta (\epsilon_i - \mu)}}, & \epsilon_i \geq \mu \\ \textcircled{II} \sum_{n_i=0}^1 e^{-\beta n_i (\epsilon_i - \mu)} = 1 + e^{-\beta (\epsilon_i - \mu)}, & \epsilon_i < \mu \end{cases}$$

$$\mathcal{Z}(T, V, \mu) = \begin{cases} \textcircled{1} \prod_i \frac{1}{1 - e^{-\beta (\epsilon_i - \mu)}}, & \epsilon_i \geq \mu \\ \textcircled{II} \prod_i (1 + e^{-\beta (\epsilon_i - \mu)}) \end{cases}$$

$$J \equiv -kT \ln \mathcal{Z} = \begin{cases} \textcircled{1} kT \sum_i \ln(1 - e^{-\beta (\epsilon_i - \mu)}) \\ \textcircled{II} -kT \sum_i \ln(1 + e^{-\beta (\epsilon_i - \mu)}) \end{cases} \rightarrow J = \pm kT \sum_i \ln(1 \pm e^{-\beta (\epsilon_i - \mu)})$$

upper signs $\rightarrow$ bosons

(J(T, V, \mu))

fugacity $\rightarrow$

$$J(T, V, z \equiv e^{\beta \mu}) = \pm kT \sum_i \ln(1 \pm z e^{-\beta \epsilon_i / kT})$$

characteristics

average occupation number: $\langle n_i \rangle = \sum_{\tilde{n}_i} \overbrace{pr(\epsilon_i, \tilde{n}_i)}^{\text{characteristics}} \cdot n_i$

we know $pr(\tilde{E}, N) = \frac{e^{-\beta \sum_i n_i (\epsilon_i - \mu)}}{\mathcal{Z}}$

$$pr(\epsilon_i, \tilde{n}_i) = \frac{1}{\mathcal{Z}} \sum_{\{n_1, \dots, n_i, \dots, n_i = \tilde{n}_i\}} e^{-\beta n_1 (\epsilon_1 - \mu)} e^{-\beta n_i (\epsilon_i - \mu)}$$

$$= \frac{1}{\mathcal{Z}} \prod_{\substack{\text{all } \epsilon\text{-levels} \\ \text{except } i\text{th}}} \left(\sum_{n_j} e^{-\beta n_j (\epsilon_j - \mu)} \right) e^{-\beta \tilde{n}_i (\epsilon_i - \mu)}$$

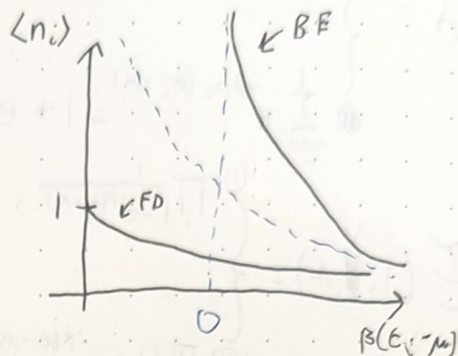
$$= \frac{e^{-\beta \tilde{n}_i (\epsilon_i - \mu)}}{\delta_i}$$

$$\Rightarrow \langle n \rangle = \sum_{\epsilon_i} \frac{e^{-\beta n_i (\epsilon_i - \mu)}}{\delta_i} \quad n_i = \underbrace{\mathcal{Z} \frac{\partial}{\partial z}}_{\text{fugacity}} \ln \delta_i \quad \text{exercise}$$

Stat Phys 2 Ideal Systems of Identical Particles 14/02

$$\langle n_i \rangle = z \frac{\partial}{\partial z} \ln \bar{\mathcal{Z}}_i$$

$$z \equiv e^{\beta \mu}$$



$$\begin{cases} J = \pm \frac{1}{\beta} \sum_i \ln(1 \mp z e^{-\beta \epsilon_i}) \\ \bar{\mathcal{Z}} = \prod_i (1 \mp z e^{-\beta \epsilon_i})^{\pm 1} \\ \bar{\mathcal{Z}} = (1 \mp z e^{-\beta \epsilon_i})^{(\mp 1)} \\ \ln \bar{\mathcal{Z}}_i = \mp \ln(1 \mp z e^{-\beta \epsilon_i}) \end{cases}$$

$$\rightarrow \langle n_i \rangle = \frac{e^{-\beta(\epsilon_i - \mu)}}{1 \mp e^{-\beta(\epsilon_i - \mu)}} \Leftrightarrow \langle n_i \rangle = \frac{1}{e^{\beta(\epsilon_i - \mu)} \mp 1}$$

Classical regime

$$\beta(\epsilon_i - \mu) \gg 1 \rightarrow \begin{cases} \langle n_i \rangle_{FD} \sim e^{-\beta(\epsilon_i - \mu)} \\ \langle n_i \rangle_{BE} \sim e^{-\beta(\epsilon_i - \mu)} \end{cases} \left. \begin{array}{l} \text{Maxwell-} \\ \text{Boltzmann} \end{array} \right\}$$

Classical $\Rightarrow \beta \epsilon_i \ll 1$ This shows that $\beta \mu \ll -1$ to compete
or equivalently $z \equiv e^{\beta \mu} \ll 1$

So $z \ll 1$ is used to identify classical regime in these systems

(BE): We assumed $\beta(\epsilon_i - \mu) > 0$, it is justified by disallowing $\langle n_i \rangle < 0$
 $\langle n_i \rangle \rightarrow \infty$ for $\beta(\epsilon_i - \mu) \rightarrow 0$

The condition $\mu < \epsilon_i$, really $\mu < \epsilon_0 \equiv$ ground state

(FD): $0 < \langle n_i \rangle < 1$

positive $\beta(\epsilon_i - \mu) \rightarrow \langle n_i \rangle \rightarrow 0$

negative $\beta(\epsilon_i - \mu) \rightarrow \langle n_i \rangle \rightarrow 1$

$\beta(\epsilon_i - \mu) \gg 1$

$$S = - \left(\frac{\partial J}{\partial T} \right)_{\mu, V}, \quad P = - \left(\frac{\partial J}{\partial V} \right)_{T, \mu}, \quad N = - \left(\frac{\partial J}{\partial \mu} \right)_{T, V}$$

$$\langle E \rangle = \frac{\partial \ln \bar{\mathcal{Z}}}{\partial \beta} \Big|_{\mu, V}, \quad \langle N \rangle = z \frac{\partial \ln \bar{\mathcal{Z}}}{\partial z} \Big|_{\mu, V}, \quad \langle E \rangle = \frac{\partial \ln \bar{\mathcal{Z}}}{\partial \beta} \Big|_{\mu, V}$$

0: Exprove these are equivalent

Stat Phys Ideal, non relativistic bosonic gas

14/02

In volume V

assume bosonic

$$\epsilon_i = \frac{\hbar^2 i^2}{2mL^2}, i=0, \pm 1, \pm 2, \dots$$

$$\mathcal{Z} = \prod_i (1 - e^{-\beta(\epsilon_i - \mu)})^{-1} \quad \mu < \epsilon_0 = 0$$

$L \sim$ characteristic scale of the vessel

$$V \sim L^3$$

$$\langle n_i \rangle = \frac{1}{e^{\beta(\epsilon_i - \mu)} - 1} \Rightarrow \langle n_0 \rangle = \frac{z}{1-z} \xrightarrow{\mu \rightarrow 0} \infty$$

$$0 < z \leq e^{\beta \mu} < 1$$

$$\ln \mathcal{Z} = - \sum_i \ln(1 - e^{-\beta(\epsilon_i - \mu)}) = - \int d\epsilon W(\epsilon) \ln(1 - z e^{-\beta \epsilon})$$

$$W(\epsilon) = \sum_i \delta(\epsilon - \epsilon_i)$$

single particle density of states

We will treat the problem approximately semi-classically

$$W(\epsilon) = \delta(\epsilon) + \sum_{i \neq 0} \delta(\epsilon - \epsilon_i) \underset{\substack{\text{dense} \\ \text{spectrum}}}{\approx} \delta(\epsilon) + (\text{deg.}) \int_{-\infty}^{\infty} d^3i \delta(\epsilon - \frac{\hbar^2 i^2}{2mL^2})$$

$$\left. \begin{aligned} p = \hbar k, \quad k_x &\approx \frac{2\pi i_x}{L} \\ k_y &\approx \frac{2\pi i_y}{L} \\ k_z &\approx \frac{2\pi i_z}{L} \end{aligned} \right\} \rightarrow \text{similar for } y, z \quad p_x = \frac{\hbar i_x}{L}$$

with this identification

$$\begin{aligned} W(\epsilon) &\approx \delta(\epsilon) + (\text{deg.}) \frac{V}{h^3} \int_{-\infty}^{\infty} d^3p \delta(\epsilon - \frac{p^2}{2m}) \\ &= \delta(\epsilon) + (\text{deg.}) \frac{V}{h^3} (4\pi) \int_0^{\infty} dp p^2 \delta(\epsilon - \frac{p^2}{2m}) \\ &\stackrel{t = \frac{p^2}{2m}}{\approx} \delta(\epsilon) + (\text{deg.}) \frac{V}{h^3} (2\pi) (2m)^{3/2} \int_0^{\infty} dt t^{1/2} \delta(\epsilon - t) \\ &= \delta(\epsilon) + (\text{deg.}) \frac{V}{h^3} (2\pi) \frac{(2m\epsilon)^{3/2}}{\epsilon} \end{aligned}$$

$$W_c(\epsilon, V, 1)$$

↑ classical

$$\ln \mathcal{Z} = - \int_{\text{all energies}} d\epsilon W(\epsilon) \ln(1 - z e^{-\beta \epsilon})$$

$$\approx - \int_0^{\infty} d\epsilon (\delta(\epsilon) + W_c(\epsilon)) \ln(1 - z e^{-\beta \epsilon})$$

$$= - \int_{-\epsilon}^{\infty} d\epsilon \delta(\epsilon) \ln(1 - z e^{-\beta \epsilon}) - \int_0^{\infty} d\epsilon W_c(\epsilon) \ln(1 - z e^{-\beta \epsilon})$$

$$= -\ln(1-z) - \int_0^{\infty} d\epsilon W_c(\epsilon) \ln(1 - z e^{-\beta \epsilon}) \Rightarrow$$

$$\ln \mathcal{Z} = -\ln(1-z) - \frac{\text{deg.} V (2\pi) (2m)^{3/2}}{h^3} \int_0^{\infty} d\epsilon \epsilon^{1/2} \ln(1 - z e^{-\beta \epsilon})$$

1462

Ideal, Non rel boson gas Stat Phys 2

$$N \equiv \text{fixed} = \langle N \rangle = \langle n_0 \rangle + \langle N_{\text{ex}} \rangle$$

for canonical ensemble, this gives $M = M(\beta, V, N)$

$$\langle N_{\text{ex}} \rangle \propto \text{Li}_{5/2}(z), \quad 0 < z < 1 \quad \text{polylogs monotonically increase}$$

Study the system in Thermo limit $\{N \gg 1\}$ with the Canonical Ensemble

$$\langle N \rangle \stackrel{\text{GCE}}{=} f(p, V, m) \stackrel{\text{C.E.}}{=} M = g(p, V, N \equiv \langle N \rangle)$$

$$\begin{array}{c} \text{fixed} \\ \downarrow \\ N = \langle n_0 \rangle + \langle N_{\text{ex}} \rangle \end{array} \begin{array}{c} \text{exchange} \\ \downarrow \end{array} \rightarrow \langle N_{\text{ex}} \rangle = N - \langle n_0 \rangle = (\text{deg}) V \left(\frac{2\pi m}{h^2 \beta} \right)^{3/2} \text{Li}_{5/2}(z)$$

$\text{Li}_{5/2}(z)$ mon increasing as z a number

$$\Rightarrow 0 < z < 1, \quad \text{Li}_{5/2}(z) < \text{Li}_{5/2}(1)$$

$$\rightarrow \langle N_{\text{ex}} \rangle \leq (\text{deg}) V \left(\frac{2\pi m}{h^2 \beta} \right)^{3/2} \text{Li}_{5/2}(1)$$

$$\text{Canonical Ensemble, for } N > N_{\text{ex}} \Rightarrow T < \underbrace{\left(\frac{h^2}{2\pi m} \left(\frac{N}{V} \right)^{2/3} \left(\frac{1}{(\text{deg}) \text{Li}_{5/2}(1)} \right)^{3/2} \right)}_{T_{\text{critical temp}}}$$

$$\text{When } T \ll T_c, \quad z = \frac{\langle n_0 \rangle}{1 - \langle n_0 \rangle} = \frac{1}{\frac{1}{\langle n_0 \rangle} - 1} \approx 1 - \frac{1}{\langle n_0 \rangle}$$

$$z \rightarrow 1 \Leftrightarrow \langle n_0 \rangle \gg 1$$

$$T/T_c \ll 1$$

$$\langle N_{\text{ex}} \rangle = N - \langle n_0 \rangle = N \left(\frac{T}{T_c} \right)^{3/2} \frac{\text{Li}_{5/2}(z)}{\text{Li}_{5/2}(1)} \stackrel{z \rightarrow 1}{\approx} N \left(\frac{T}{T_c} \right)^{3/2}$$

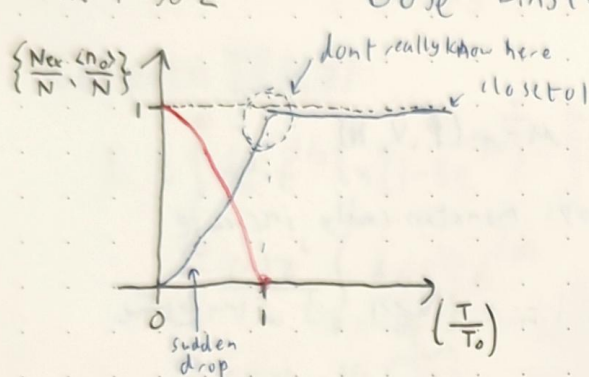
$$\langle n_0 \rangle \approx N \left(1 - \left(\frac{T}{T_c} \right)^{3/2} \right), \quad \frac{T}{T_c} \ll 1 \text{ or } z \rightarrow 1$$

B-E condensation, low temperatures $\left. \begin{array}{l} z \rightarrow 1 \\ \text{bound on } N_{\text{ex}} \end{array} \right\} \text{Quantum Physics}$

$$\text{we expect classical results: } N_{\text{ex}} = N \left(\frac{T}{T_c} \right)^{3/2} \frac{\text{Li}_{5/2}(z)}{\text{Li}_{5/2}(1)} \approx N \left(\frac{T}{T_c} \right)^{3/2} \frac{z}{\text{Li}_{5/2}(1)}$$

$$\text{Li}_{5/2}(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^{5/2}} \approx z$$

Note that this result is constant with $\langle n_0 \rangle \ll 1, N \approx N_{\text{ex}} \Rightarrow \left(\frac{T}{T_c} \right)^{3/2} \approx \frac{1}{z} \gg 1$



At $T = T_c$ we see a new phase.

BE condensation is a phase transition

Equation of State

$$\frac{z}{1-z} = \langle n_0 \rangle$$

$$PV = NkT \text{ classically, } = \frac{2}{3} \langle E \rangle$$

$$J = -kT \ln \Xi = -P \cdot V, \quad \langle E \rangle = - \left. \frac{\partial \ln \Xi}{\partial \beta} \right|_{V, z} \left. \vphantom{\frac{\partial \ln \Xi}{\partial \beta}} \right\} \Rightarrow P \cdot V = \frac{2}{3} \langle E \rangle - kT \ln(1-z)$$

$$= \frac{2}{3} \langle E \rangle + kT \ln(1 + \langle n_0 \rangle)$$

Exercise

probability for

to have

What is the average no of particles with energies between E & $E + dE$

$$\langle n(E) \rangle$$

exact average occupation number

$$\langle N_{ex} \rangle = a V \beta^{-3/2} \text{Li}_{3/2}(z)$$

$$d\langle N \rangle = \langle n(E) \rangle W(1, E) dE$$

$$\int_0^\infty d\langle N \rangle \approx \langle N_{ex} \rangle = a V \beta^{-3/2} \text{Li}_{3/2}(z)$$

$$P(E) dE = \frac{\langle n(E) \rangle}{\langle N_{ex} \rangle} W_c(1, E) dE \quad (\text{check: } \int_0^\infty P(E) dE = 1)$$

$$\Rightarrow P(E) dE = \frac{\langle n(E) \rangle}{\text{Li}_{3/2}(z)} \frac{z}{\sqrt{\pi}} E^{1/2} \beta^{3/2} dE$$

Exercise: What is the average energy level of the system?

$$\langle E \rangle = \int_0^\infty dE E P(E) = \frac{z}{\sqrt{\pi}} \beta^{3/2} \frac{1}{\text{Li}_{3/2}(z)} \underbrace{\int_0^\infty dE \langle n(E) \rangle E^{3/2}}_I$$

$$\text{Compute I: } \int_0^\infty dE \frac{E^{3/2}}{e^{\beta(E-\mu)} - 1} = \int_0^\infty \frac{dE}{\frac{e^{\beta E}}{z} - 1} E^{3/2} = \frac{z}{\beta^{5/2}} \int_0^\infty \frac{t^{3/2}}{e^t - 1} dt = \beta^{-5/2} \frac{\Gamma(5/2)}{\text{Li}_{5/2}(z)}$$

Stat Phys 2 Ideal N-R Bosonic Gas

21/02

$$\rightarrow \langle E \rangle = \left(\frac{2}{\sqrt{\pi}} \Gamma\left(\frac{5}{2}\right) \right) \frac{Li_{5/2}(z)}{Li_{3/2}(z)} \beta^{-1}$$

~~Exercise~~ Compute $C_{v,N}$ for $T \approx T_c$ ($T \rightarrow T_c$) for a N-R Bosonic Gas

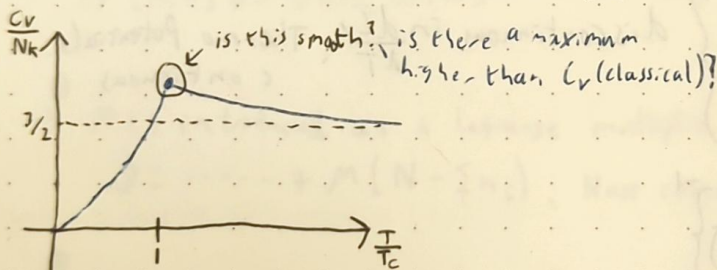
$$C_v = \left. \frac{\partial \langle E \rangle}{\partial T} \right|_{v,N} \quad \text{For } T \approx T_c, z \approx N, N \approx N_{ex} \approx a V \rho_c^{-3/2} Li_{3/2}(1)$$

$$\xrightarrow{\text{sub into } \langle E \rangle} C_v = \frac{15}{4} \frac{Li_{5/2}(1)}{Li_{3/2}(1)} \left(\frac{T}{T_c} \right)^{3/2} \propto 1.925 \left(\frac{T}{T_c} \right)^{3/2}$$

* Compute C_v for $T \gg T_c$ ($z \ll 1$) to show $C_v = \frac{3}{2} N k_B$

$$z \ll 1, Li_{5/2}(z) \approx z \quad T \gg T_c, N \approx N_{ex} \approx a V \rho_c^{-3/2} z$$

$$\xrightarrow{\text{sub into } \langle E \rangle} \langle E \rangle \approx \frac{3}{2} N \beta^{-1} \Rightarrow C_v \approx \frac{3}{2} k_B N$$



What happens to C_v as $T \rightarrow T_c^+$

$$C_{v,N} = \left. \frac{\partial \langle E \rangle}{\partial T} \right|_{v,N} = \frac{-15}{4} a \rho_c^{-3/2} Li_{5/2}(z) + \frac{3}{2} a \rho_c^{-5/2} Li_{3/2}(z) \frac{d \ln z}{dT} = \left. \frac{\partial \langle E \rangle}{\partial T} \right|_{v,N}$$

$$\text{start } N \approx N_{ex} \approx a k^{3/2} T^{3/2} Li_{3/2}(z) \text{ for } T > T_c$$

$$z \frac{d Li_{3/2}(z)}{dz} = Li_{1/2}(z)$$

$$\text{canonical ensemble, } N \text{ fixed} \rightarrow \frac{dN}{dT} = 0 \Leftrightarrow a k^{3/2} \frac{3}{2} T^{1/2} Li_{3/2}(z) + a k^{3/2} T^{3/2} \frac{Li_{3/2}(0)}{z} \left(\frac{dz}{dT} \right) = 0$$

$$\Rightarrow \left(\frac{dz}{dT} \right)_{v,N} \approx - \frac{3}{2T} \left(\frac{z Li_{3/2}(z)}{Li_{1/2}(z)} \right) \text{ how does this behave for } z \rightarrow 1 \quad \int_0^\infty dx \frac{x^{-1/2}}{(e^x - 1)}$$

For $z \rightarrow 1$, the integral in the polylog diverges close to $x \approx 0$, so low values of x give the dominant contribution so $z \sim 1$

$$\{ \quad L_{1/2}(z) \stackrel{z \rightarrow 1}{\sim} \frac{1}{\Gamma(1/2)} \int_0^{\infty} \frac{dx x^{-1/2}}{(x+1)z - 1} + \text{"finite terms"} = \frac{z}{\Gamma(1/2)} \int_0^1 \frac{x^{-1/2}}{(1-z+x)} + \text{"finite terms"} \\ \stackrel{u \equiv \sqrt{x}}{=} \frac{z z(1-z)^{-1/2}}{\Gamma(1/2)} \int_0^1 \frac{du}{1+u^2} + \text{"finite terms"}$$

$$\rightarrow L_{1/2}(z) \propto_{z \rightarrow 1} z(1-z)^{-1/2} \quad \text{and} \quad \frac{dz}{dT} \underset{z \rightarrow 1}{\approx} -\frac{z}{2T} \sqrt{1-z} \frac{N}{ak^{3/2} T^{3/2}} \stackrel{\text{known}}{\approx} A \frac{\sqrt{1-z(T)}}{T^{5/2}}$$

$\rightarrow 1^{\text{st}}$ order ODE for $z = z(T)$, $T \rightarrow T_c^+$

$$z(T_c) = 1 \quad \text{leads to} \quad z(T) \underset{\text{known}}{\approx} 1 - b \left(\frac{T - T_c}{T_c} \right)^2$$

$$\xrightarrow{\text{substit } C_v} \frac{C_v}{kN} \underset{T \rightarrow T_c^+}{\approx} \frac{15}{4} \frac{L_{3/2}(z)}{L_{3/2}(1)} \left(\frac{T}{T_c} \right)^{3/2} - \frac{9}{4} \cdot \# \cdot \left(\frac{T - T_c}{T_c} \right)^2$$

So: ① C_v smoothly approaches T_c as $T \rightarrow T_c^+$

$$\left. \begin{aligned} \text{② } \frac{dC_v}{dT} \Big|_{T \rightarrow T_c^+} &= \left(\dots \right)_{1^{\text{st}} \text{ terms}} + \# \\ \frac{dC_v}{dT} \Big|_{T \rightarrow T_c^-} &= \left(\dots \right)_{1^{\text{st}} \text{ terms}} \end{aligned} \right\} \begin{array}{l} \text{discontinuous in } \frac{dC_v}{dT}, \text{ Thermo Potentials} \\ \text{continuous} \end{array}$$

$$z = \left(1 + \frac{1}{L_{3/2}} \right)_{T \rightarrow T_c}^{-1} \left\{ 1 + \frac{1}{N \left(1 - \left(\frac{T}{T_c} \right)^{3/2} \right)} \right\}^{-1}$$

$$\frac{dz}{dT} \Big|_{N, N} \underset{T \rightarrow T_c}{\sim} \cancel{O(1/N)} O\left(\frac{1}{N}\right) \quad \frac{dz}{dT} \Big|_{N, N} \underset{T \rightarrow T_c}{\sim} O(1)$$

The discontinuity arises because we consider $N \gg 1$ & study the system in the leading $N \rightarrow \infty$ approx. (Thermo Limit)

Comment about what is and is not B-E condensate

Consider the gas, $\epsilon_1 - \epsilon_0 \approx \frac{\hbar^2}{2mV^{2/3}}$, so $kT < \epsilon_1 - \epsilon_0$, then particles will both $i=0$ state but this is not BEC.

Macroscopic $\rightarrow \frac{\hbar^2}{2mV^{2/3}} \ll kT \ll kT_c$ & particles will still populate the ground state much more

Stat Phys 2 Photon Gas

21/02

Semi-classical treatment of "blackbody" radiation aka "photon gas"

A system of bosonic particles, which are free and can be treated with the methods discussed so far

* Relativistic Particles ($m=0$)

** Number of Photons is not conserved $\Rightarrow \mu=0$

* $\Rightarrow N$ not conserved $\Rightarrow$ no work associated with change of N $\mu=0$

$$\hat{\rho} = \frac{e^{-\beta \hat{H}}}{\mathcal{Z}} \quad \text{density matrix} \\ \text{Canonical ensemble}$$

generality

$$\mathcal{Z} = e^{\beta \mu} = 1 \text{ bec}$$

$$\hat{\rho} = \frac{e^{-\beta(\hat{H} - \mu \hat{N})}}{\mathcal{Z}} \quad \text{grand canonical ensemble}$$

$$[\hat{\rho}, \hat{H}] = 0$$

$$\text{conserved} \Leftrightarrow [\hat{N}, \hat{H}] = 0$$

$$\text{if } [\hat{N}, \hat{H}] \neq 0 \text{ then } [\hat{\rho}, \hat{H}] \neq 0 \text{ unless } \mu=0$$

* $\Rightarrow \mu$ is introduced as a Lagrange multiplier in the relevant thermodynamic potential

$$\Phi = \dots + \mu(N - \sum n_i) \quad N \text{ not conserved so no reason to define } \mu$$

* $\Rightarrow$ If N is not conserved, then F or G should not depend on $N \Rightarrow \mu \propto \left(\frac{\partial F}{\partial N}\right)_{TV} = 0$

relativistic means
density of states is what matters

$$W_c(1, \epsilon) \quad \ln(\mathcal{Z}) = \int d\epsilon W_c(1, \epsilon) \ln(1 - \mathcal{Z} e^{-\beta \epsilon})$$

$$W_c(1, p) dp = (\text{deg}) 4\pi \frac{V}{h^3} p^2 dp = (\text{deg}) \frac{4\pi V}{h^3} (\epsilon^2 - m^2) \cdot \frac{2\epsilon d\epsilon}{2\sqrt{\epsilon^2 - m^2}} = (\text{deg}) \frac{4\pi V}{h^3} \sqrt{\epsilon^2 - m^2} \epsilon d\epsilon$$

set $c=1 \rightarrow \epsilon = \sqrt{p^2 + m^2}$

For photons: $(\text{deg}) = 2$ (2 polarisations) Take as given for this module
 $m=0$

$$\text{then } W_c(1, \epsilon) d\epsilon = \frac{8\pi V}{c^3 h^3} \epsilon^2 d\epsilon$$

$$\rightarrow \ln \mathcal{Z} = - \int d\epsilon \frac{8\pi V}{c^3 h^3} \epsilon^2 \ln(1 - e^{-\beta \epsilon})$$

Stat Phys 2 Photon Gas

27/02

$$\ln \tilde{Z} = - \int_0^\infty d\epsilon \left(\frac{8\pi V}{c^3 h^3} \right) \epsilon^2 \ln(1 - e^{-\beta \epsilon}) = - \frac{8\pi V}{c^3 h^3} I(\beta)$$

integration by parts

$$I(\beta) = - \int_0^\infty d\epsilon \frac{\epsilon^3}{3} \frac{1}{1 - e^{-\beta \epsilon}} \stackrel{x = \beta \epsilon}{=} - \frac{\beta^3}{3} \int_0^\infty dx \frac{x^3}{e^x - 1} = - \frac{\beta^3}{3} \frac{\pi^4}{15}$$

$$\int_0^\infty dx \frac{x^n e^{-x}}{1 - e^{-x}} = \sum_{n=0}^\infty \int_0^\infty dx e^{-(n+1)x} x^n = \sum_{n=0}^\infty \frac{1}{(n+1)^{n+1}} = \sum_{n=1}^\infty \frac{1}{n^{n+1}}$$

$$\Rightarrow \ln \tilde{Z} = \frac{8\pi^5 V}{15 c^3 h^3} \beta^{-3} \Rightarrow J = -kT \ln \tilde{Z} = -PV \langle E \rangle - \frac{\partial \ln \tilde{Z}}{\partial \beta}$$

$$\langle E \rangle = - \frac{\partial \ln \tilde{Z}}{\partial \beta} = - \frac{8\pi^5 V}{15 c^3 h^3} (-3) \beta^{-4} = \frac{8\pi^5}{15 c^3 h^3} \beta^{-4} V$$

$$\Rightarrow \langle E \rangle = \left(\frac{\langle E \rangle}{V} \right) = \frac{8\pi^5}{15 c^3 h^3} (kT)^4 \propto T^4$$

related to the dimensionality of spacetime here

$$J = -P \cdot V \Rightarrow PV = \frac{1}{3} \langle E \rangle V \Rightarrow P = \frac{1}{3} \langle E \rangle$$

total average energy density

$$S = \frac{\langle E \rangle - J}{T} = \frac{\langle E \rangle - (-PV)}{T} = \frac{\langle E \rangle + PV}{T}$$

$$\Rightarrow S = 4 \cdot \frac{PV}{T} = \frac{4}{3} V \frac{\langle E \rangle}{T} \Rightarrow S \propto \frac{4}{3} VT^3$$

What will happen if the radiation undergoes a reversible adiabatic change
 $VT^3 = \text{const}$ $PV^{4/3} = \text{const}$

$PV^\gamma = \text{const}$, classical gas $\Rightarrow \gamma \sim \frac{C_p}{C_v}$, Not true in QM

$$C_v = T \left(\frac{\partial S}{\partial T} \right)_V = 3S \sim T^3, \quad \frac{C_p}{C_v} = \frac{\left(\frac{\partial V}{\partial P} \right)_T}{\left(\frac{\partial V}{\partial P} \right)_S} = \frac{1}{\left(\frac{\partial P}{\partial V} \right)_T} \cdot \frac{1}{\beta^2} \rightarrow \infty$$

Recap: "Black body" $\leftrightarrow$ gas of photons, bosons, c

1) Relativistic, $c \rightarrow U_c(1, \epsilon)$

2) N not conserved, $M=0$

grand canonical $\sim$ canonical

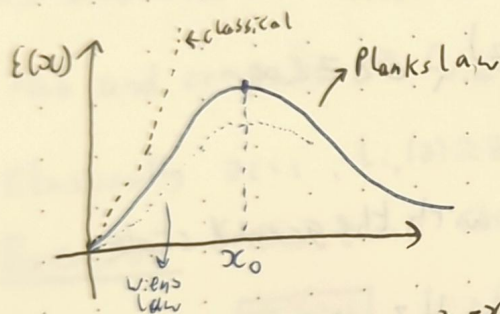
$$\langle E \rangle = \frac{\langle E \rangle}{V} = \frac{8\pi^5}{15 c^3 h^3} \beta^{-4}$$

We can define, since $\langle \epsilon \rangle = \int_0^\infty d\epsilon \frac{W_c(l, \epsilon)}{V} \epsilon \langle n(\epsilon) \rangle$

$$\epsilon(\epsilon) \stackrel{de}{=} \frac{W_c(l, \epsilon)}{V} \epsilon \langle n(\epsilon) \rangle_{\mu=0} d\epsilon$$

express in terms of frequency ω using $\omega \equiv \frac{\epsilon}{\hbar}$

$$\Rightarrow \epsilon(\omega) d\omega \stackrel{h \cdot \omega}{=} \frac{\hbar \cdot \omega^3}{\pi^2 c^3} \frac{1}{e^{\beta \hbar \omega} - 1} d\omega \Rightarrow \epsilon(x) dx \propto \frac{x^3}{e^x - 1} dx$$



$$x_0 \equiv \beta_0 \hbar \omega_0 \Rightarrow \omega_0 = \hbar \frac{1}{\beta_0} \frac{k T_0}{\hbar}$$

This determines the max of the distribution determined by $\frac{d\epsilon(x)}{dx} \Big|_{x_0} = 0$

$$\text{QM} \quad x \gg 1 \rightarrow \epsilon(x) \approx x^3 e^{-x}$$

$$\text{CM} \quad x \ll 1 \rightarrow \epsilon(x) \approx x^2 \rightarrow \text{Ultraviolet Catastrophe!}$$

Stat Phys 2 NR Fermionic Gas

2802
 $0 < z < \infty$

$$\ln \tilde{Z} = \sum_i \ln(1 + z e^{-\beta \epsilon_i}) = \ln(1+z) + \int d\epsilon W(\epsilon) \ln(1 + z e^{-\beta \epsilon})$$

$$W(\epsilon) d\epsilon = (\text{deg}) \frac{V}{h^3} (2m)^{3/2} 2\pi \epsilon^{1/2} d\epsilon$$

$$\Rightarrow \ln \tilde{Z} = \ln(1+z) - (\dots) \int_0^{\infty} \frac{dx x^{3/2}}{1 + \frac{e^{\beta \mu}}{z}} = \text{Li}_{5/2}(-z)$$

$$Li_r(-z) = -\frac{1}{\Gamma(r)} \int_0^{\infty} \frac{dt t^{r-1}}{1 + \frac{e^t}{z}}$$

↑
converges for $z \geq -1$

$$\rightarrow \ln \tilde{Z} = \ln(1+z) - (\text{deg}) \left(\frac{2\pi m}{h^2} \right)^{3/2} V \beta^{-3/2} Li_{5/2}(-z), \quad 0 < z < \infty$$

• there is no bound on z & hence on $Li_{5/2}$
 thus there is no special physics associated with the ground state

• The polylog term dominates for both large & small $z \Rightarrow$ we can neglect the ground state contribution

$$\rightarrow \ln \tilde{Z} = -(\text{deg}) \left(\frac{2\pi m}{h^2} \right)^{3/2} V \beta^{-3/2} Li_{5/2}(-z)$$

$$\langle N \rangle = z \frac{\partial \ln \tilde{Z}}{\partial z} = -(\text{deg}) \left(\frac{2\pi m}{h^2} \right)^{3/2} V \beta^{-3/2} Li_{3/2}(-z) \quad \text{A \& R \& W}$$

$$Li_r(-z) \underset{z \ll 1}{\approx} -z \left\{ 1 - \frac{z}{2} + \dots \right\}, \quad Li_r(-z) \underset{z \gg 1}{\approx} -\frac{(\ln z)^r}{\Gamma(r+1)} \left\{ 1 + \frac{\pi^2}{6} \frac{\Gamma(r-1)}{(\ln z)^2} \right\}$$

$$\langle E \rangle = -\frac{\partial \ln \tilde{Z}}{\partial \beta} \Big|_{\mu, V} = -\frac{3}{2} (\text{deg}) \left(\frac{2\pi m}{h^2} \right)^{3/2} V \beta^{-5/2} Li_{5/2}(-z) = \frac{3}{2} P \cdot V$$

$$\text{but also } \langle E \rangle = \frac{3}{2} P \langle N \rangle \frac{Li_{5/2}(-z)}{Li_{3/2}(-z)} = \begin{cases} z \gg 1 \rightarrow \frac{\langle N \rangle}{\beta} \ln z & \text{"QM"} \\ z \ll 1 \rightarrow \frac{3}{2} kT \langle N \rangle & \text{"classical"} \end{cases}$$

$$F = \langle E \rangle - T S = \mu \langle N \rangle + J$$

$$\ln z = \beta \mu \rightarrow \mu = \langle N \rangle kT \left\{ \ln z - \frac{Li_{5/2}(-z)}{Li_{3/2}(-z)} \right\}$$

$$S = \frac{1}{T} (\langle E \rangle - F) = \langle N \rangle k \left\{ \frac{5}{2} \frac{Li_{5/2}(-z)}{Li_{3/2}(-z)} - \ln z \right\}$$

$$C_{V,N} = T \left(\frac{\partial S}{\partial T} \right)_{V,N} = T \left(\frac{\partial S}{\partial z} \right)_{V,N} \cdot \left(\frac{\partial z}{\partial T} \right)_{V,N}$$

$$\left(\frac{\partial z}{\partial T} \right)_{V,N} = ?$$

Stat Phys 2 N-R Fermionic Gas

28/02

$$\left(\frac{\partial Z}{\partial T}\right)_{V,N} = ? \quad \text{Set } N = \text{fixed} = \langle N \rangle \Rightarrow \left(\frac{\partial N}{\partial T}\right)_{V,N} = 0$$

$$\Rightarrow 0 = (\text{deg}) V \left(\frac{2\pi m k}{h^2}\right)^{3/2} \left\{ -\frac{3}{2} T^{1/2} \text{Li}_{3/2}(-z) - T \frac{\partial}{\partial T} \text{Li}_{5/2}(-z) \right\}$$

$$\Rightarrow -\frac{3}{2} T^{1/2} \text{Li}_{3/2}(-z) = -T^{3/2} \frac{\text{Li}_{3/2}(-z)}{(-z)} \left(\frac{\partial z}{\partial T}\right)_{V,N}$$

$$\Rightarrow \frac{1}{z} \left(\frac{\partial z}{\partial T}\right)_{V,N} = -\frac{3}{2T} \frac{\text{Li}_{3/2}(-z)}{\text{Li}_{1/2}(-z)}$$

$$\text{sub and compute} \rightarrow C_{V,N} = -\frac{9}{4} N k \frac{\text{Li}_{3/2}(-z)}{\text{Li}_{1/2}(-z)} + \frac{15}{4} N k \frac{\text{Li}_{5/2}(-z)}{\text{Li}_{1/2}(-z)}$$

$$\text{classically } z \ll 1, \text{Li}_\nu(z) \approx z \rightarrow C_{V,N} \approx \frac{3}{2} N k$$

Exercise Show that classical $\rightarrow S$ behaves as expected

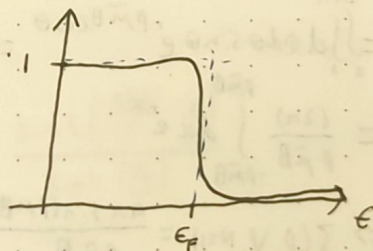
Not doing exercise $\boxed{z \gg 1} \quad \text{Li}_{3/2}(-z) \underset{z \gg 1}{\approx} -\frac{(\ln z)^{3/2}}{\Gamma(5/2)} = -\frac{\langle N \rangle}{V} \left[\left(\frac{h^2}{2\pi m}\right)^{1/2} \frac{1}{(\text{deg})} \right] \beta^{3/2}$
solving in terms of $\langle N \rangle$

$$\Rightarrow \mu(z \gg 1) \approx \# \left(\frac{N}{V}\right)^{2/3} \equiv \text{fixed independent of } T$$

μ is a constant at small temperatures

$$\text{As } T \rightarrow 0 \quad u(T \rightarrow 0) \approx \epsilon_F \quad \text{fermi energy}$$

$$\rightarrow \langle n(\epsilon) \rangle = \frac{1}{e^{\beta(\epsilon - \mu)} + 1} \underset{T \rightarrow 0}{=} \begin{cases} 1 & \epsilon < \epsilon_F \\ 0 & \epsilon > \epsilon_F \end{cases}$$



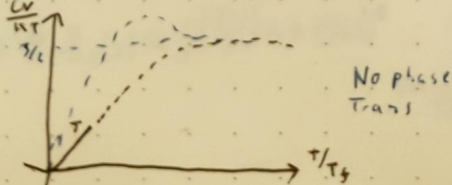
$$\left. \begin{aligned} N &= \int_0^\infty d\epsilon W_c(1, \epsilon) \langle n(\epsilon) \rangle \underset{T \gg 1}{=} \int_0^{\epsilon_F} d\epsilon W_c(1, \epsilon) = \# \epsilon_F^{3/2} \\ \langle E \rangle &\underset{T \gg 0}{=} \int_0^\infty d\epsilon \epsilon \langle n(\epsilon) \rangle W_c(1, \epsilon) = \int_0^{\epsilon_F} d\epsilon \epsilon W_c(1, \epsilon) = \# \epsilon_F^{5/2} \end{aligned} \right\} \Rightarrow \frac{3}{5} N \epsilon_F = \langle E \rangle$$

let's study C_V for $\beta \gg 1$ ($z \gg 1$)

Subbing in leading polylog terms sends C_V to 0 not good
 Keep more terms in polylog expansion

$$\text{Doing so} \Rightarrow C_V \underset{T \rightarrow 0}{\approx} \frac{\pi^2}{2} N k \left(\frac{kT}{\epsilon_F}\right) \propto T$$

unlike bosons which are T^3 , so we can tell what we have with this behaviour
 $T_3 \sim \frac{\epsilon_F}{k}$

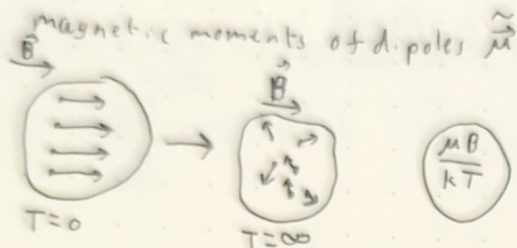


Statphys 2 Statistics of Paramagnetism

13/03

N magnetic dipoles interacting with a magnetic $\vec{B}$

Look classical first but the exam will be focus on quantum



Completely localised dipoles

Classical Treatment

assume all dipoles have $|\vec{\mu}| = \mu$

$$E = \sum_{i=1}^N E_i = - \sum_{i=1}^N \vec{\mu}_i \cdot \vec{B} = - \sum_{i=1}^N \mu B \cos \theta_i$$

Thermo limit: $Z = \prod_{i=1}^N \mathcal{Z}_i = \mathcal{Z}^N$ because there is no interaction between the dipoles

$$\mathcal{Z} = \mathcal{Z}(T, V, 1) = \int_{\Omega} e^{-\beta \vec{\mu} \cdot \vec{B} \cos \theta} d\Omega$$

not really a sum, θ, ϕ continuous

$$= \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta e^{-\beta \mu B \cos \theta} = 2\pi \int_{-1}^1 d(\cos \theta) e^{+\beta \mu B \cos \theta}$$

$$x = \beta \mu B \cos \theta$$

$$\rightarrow = \frac{(2\pi)}{\beta \mu B} \int_{-\beta \mu B}^{+\beta \mu B} dx e^{+x}$$

$$\Rightarrow \mathcal{Z}(\beta, V, N=1) = \frac{4\pi \mu B \sinh(\beta \mu B)}{\beta \mu B} \Rightarrow Z = \mathcal{Z}^N = \left(\frac{4\pi \mu B \sinh(\beta \mu B)}{\beta \mu B} \right)^N$$

$$F = -kTN \ln \mathcal{Z}$$

$$\langle \vec{\mu} \cos \theta \rangle \equiv \text{average dipole moment of each particle} = \langle M \rangle$$

$$N \langle \vec{\mu} \cos \theta \rangle \equiv \text{all of the system}$$

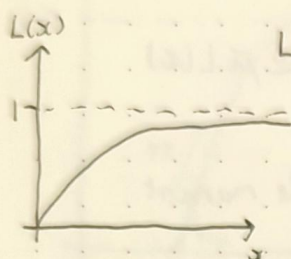
$$\langle \mu \cos \theta \rangle = \frac{\int \int d\phi d\theta \sin \theta \mu \cos \theta e^{-\beta \mu B \cos \theta}}{\int \int d\phi d\theta \sin \theta e^{-\beta \mu B \cos \theta}} = \frac{1}{\mathcal{Z}} \left(\frac{\partial \mathcal{Z}}{\partial \beta} \right) = \frac{1}{\mathcal{Z}} \left(\frac{\partial \ln \mathcal{Z}}{\partial \beta} \right)_{V, N}$$

$$\Rightarrow -\frac{1}{N} \left(\frac{\partial F}{\partial B} \right) = \langle M \rangle$$

$$\chi_T \equiv \text{isothermal mag. susceptibility} = \left(\frac{\partial \langle M \rangle}{\partial B} \right)_{\beta, V, N}$$

So let us evaluate: $\langle M \rangle = \dots = -\frac{1}{\beta B} + \tilde{M} \coth(\beta \tilde{M} B)$

$$= \tilde{M} \left(\underbrace{-\frac{1}{\beta \tilde{M} B}}_{L(x)} + \coth(\beta \tilde{M} B) \right)$$



$$L(x) = \coth(x) - \frac{1}{x}$$

$$L(x) \quad x \equiv \beta \tilde{M} B$$

1) $x \gg 1, \tilde{M} B \gg kT$

$$L(x) \rightarrow 1$$

$$\langle M \rangle = \tilde{M}$$

2) $x \ll 1, \tilde{M} B \ll kT$

$$L(x) \approx \frac{x}{3} - \frac{x^3}{45} + \mathcal{O}(x^5)$$

$$\langle M \rangle \approx \frac{M B}{3 k T} \rightarrow \langle M \rangle \propto \frac{1}{T} \quad \leftarrow \text{Curie law para mag}$$

$$\hat{H} = -\vec{\mu} \cdot \vec{B} \quad \text{quantisation}$$

$$\vec{\mu} = g \left(\frac{q}{2mc} \right) \cdot \left\{ \begin{array}{l} \uparrow \\ \text{combination } \vec{J} \end{array} \right.$$

↑
magnetic ratio

$$\vec{\mu} = -g \tilde{\mu}_0 \cdot \vec{m} \cdot \vec{B}$$

↑
quantisation no $m = -J, -J+1, \dots, J$

$$\tilde{\mu}_0 = \frac{q \hbar}{2mc}$$

$$\mathcal{Z}(\beta, V, B, N=1) = \sum_{m=-J}^J e^{\beta g \tilde{\mu}_0 B m}$$

$$\Rightarrow \mathcal{Z} = \sum_{m=-J}^J e^{-\frac{m x}{J}} = e^{-x} \left(\frac{e^{\frac{2J+1}{J} x} - 1}{e^{\frac{x}{J}} - 1} \right) = \frac{\sinh\left(\frac{2J+1}{J} x\right)}{\sinh\left(\frac{x}{J}\right)}$$

$x = \beta g \mu_0 B J$

$$Z = \mathcal{Z}^N, F = -kT \ln Z \rightarrow \langle M \rangle = \frac{1}{\beta} \left(\frac{\partial \ln \mathcal{Z}}{\partial B} \right)_{\beta, V}, \chi_T = \left(\frac{\partial \langle M \rangle}{\partial B} \right)$$

$$\langle M \rangle = \frac{1}{\beta} \left(\frac{\partial \ln \mathcal{Z}}{\partial B} \right)_{\beta, V} = g \mu_0 J \left\{ \frac{2J+1}{2J} \coth \left[x \left(\frac{2J+1}{2J} \right) \right] - \frac{1}{2J} \coth \left(\frac{x}{2J} \right) \right\}$$

$\mathcal{B}_J(x)$: Brillouin function order J

① $x \gg 1$ (Strong magnetic vs Temp)

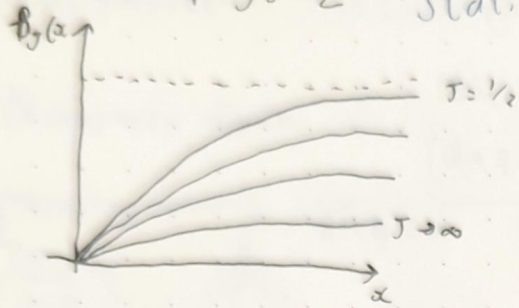
$$\mathcal{B}_J(x) \approx 1 \quad \forall J$$

magnetic saturation

② $x \ll 1$ (weak magnetic vs Temp)

$$\mathcal{B}_J(x) \approx \frac{1}{J} \left(\frac{J+1}{J} \right) x + \mathcal{O}(x^3) \Rightarrow \langle M \rangle \approx \frac{g^2 \tilde{\mu}_0^2 J(J+1) H}{3k}$$

only leading term



When do we get the classical Result?
 $T \rightarrow \infty, g \rightarrow 0$ ($\epsilon_m = g \mu_B \tilde{\mu}_0 B$)

$$\left. \begin{aligned} &B_z(x) \text{ indep of } J \\ &\coth\left(\frac{x}{2T}\right) \approx \frac{2T}{x} \end{aligned} \right\} \rightarrow \langle M \rangle \approx \tilde{\mu}_0 L(x)$$

$\rightarrow g J \tilde{\mu}_0 \equiv \tilde{\mu}$ classical dipole moment

Now check when we fix $J = \frac{1}{2}$
 $\Rightarrow \epsilon_i = \begin{cases} -\tilde{\mu}_0 B \equiv -\epsilon_0 \\ +\tilde{\mu}_0 B \equiv \epsilon_0 \end{cases}$

Thermal limit, canonical ensemble

$$Z = \delta^N = \left(\frac{2 \cosh(2x)}{2} \right)^N = 2^N \cosh^N(x), \quad x = \beta \tilde{\mu}_0 B = \beta \epsilon_0$$

$$F = -kT \ln Z = -kTN \ln [2 \cosh(\beta \epsilon_0)]$$

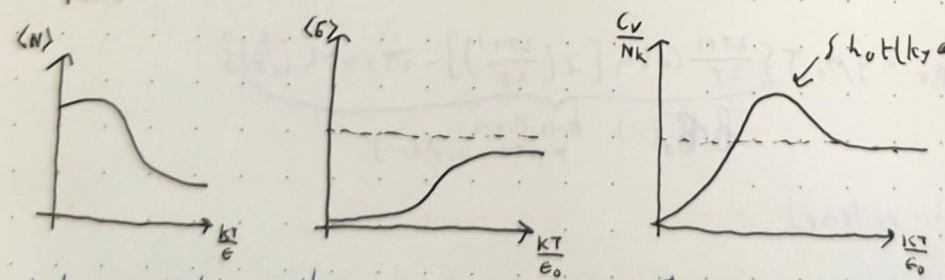
$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,B,N} = Nk \{ \ln [2 \cosh(\beta \epsilon_0)] - \beta \epsilon_0 \tanh(\beta \epsilon_0) \}$$

$$\langle E \rangle = F + TS = -\frac{\partial \ln Z}{\partial \beta} = -N \epsilon_0 \tanh(\beta \epsilon_0) = -N \langle M \rangle B$$

$$\langle M \rangle = -\frac{1}{N} \left(\frac{\partial F}{\partial B} \right)_{T,V,N} = \tilde{\mu}_0 \tanh(\beta \epsilon_0) \quad C_V = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_{V,H,N} = \frac{Nk(\beta \epsilon_0)^2}{\cosh^2(\beta \epsilon_0)}$$

$$S \underset{\beta \epsilon_0 \ll 1}{\approx} Nk [\ln(2) - (\beta \epsilon_0)^2] \approx Nk \ln 2 \quad \Rightarrow \Omega = 2^N, kT \gg \epsilon_0 \text{ disorder}$$

$$S \underset{\beta \epsilon_0 \gg 1}{\approx} Nk [\beta \epsilon_0 - \beta \epsilon_0] \approx 0 \quad \Rightarrow \Omega = 0, kT \ll \epsilon_0 \text{ order all in line with } B \text{ field}$$



Why can't they all be ordered opposite to B 's field
 Solve for T from $\langle E \rangle \Rightarrow \frac{1}{T} = \frac{k}{2\epsilon_0} \ln \left(\frac{1 - \frac{\langle E \rangle}{N\epsilon_0}}{1 + \frac{\langle E \rangle}{N\epsilon_0}} \right)$
 $\Rightarrow T < 0$ is a characteristic feature of systems with upper bound on energy

$$S = N k \ln 2$$

$\langle E \rangle \Rightarrow 0$

↖ half positive
half negative

- when $\langle E \rangle \rightarrow 0^+$ what happens

$$\frac{\partial S}{\partial E} \sim \frac{1}{T} < 0$$

In this case $\exists T$ s.t. $kT \gg \frac{E}{N}$

Canonical Ensemble: $Z \approx (\sum_n e^{-\beta E_n})^N$ neglecting interactions for high T

$$\beta E_n \ll 1 \Rightarrow Z \simeq \left(\sum_n \left(1 - \beta E_n + \frac{1}{2} \beta^2 E_n^2 \right) \right)^N$$

$$\ln Z \approx N \ln \left[\sum_n 1 - \sum_n \rho \epsilon_n + \frac{1}{2} \sum_n \rho^2 \epsilon_n^2 \right]$$

To proceed we set $\sum_n 1 \approx g$, e.g. all possible orientations in a magnetic field,
all possible polarizations etc.

$$\sum p_n \epsilon_n \approx g \beta \langle \epsilon_n \rangle$$

$$\sum_n \beta^2 \epsilon_n^2 \approx g \beta^2 \langle \epsilon_n^2 \rangle$$

$$\text{Sub into } \ln Z \Rightarrow \ln Z \simeq N \ln \left[g(1 - \rho \langle \epsilon_n \rangle + \frac{1}{2} \rho^2 \langle \epsilon_n^2 \rangle) \right]$$

$$\overline{P(\epsilon_n < \epsilon_1)} N \left\{ \ln g - \left[P(\epsilon_n) - \frac{1}{2} P^2(\epsilon_n) \right] - \frac{1}{2} \left[P(\epsilon_n) - \frac{1}{2} P^2(\epsilon_n) \right]^2 + \dots \right\}$$

$$\ln Z \simeq N [\ln g - \beta \langle \epsilon_n \rangle + \frac{1}{2} \beta^2 (\langle \epsilon_n^2 \rangle - \langle \epsilon_n \rangle^2)]$$

$$F = -kT \ln Z \simeq -NkT \ln g + N \langle E_n \rangle - \frac{N}{2kT} \langle (E - \langle E_n \rangle)^2 \rangle$$

$$S \simeq N k \ln g - \frac{Nk}{2} \beta^2 \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \quad \langle E \rangle \simeq N \langle \epsilon_n \rangle - N \beta \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle$$

$C_v \approx Nk\beta^2 \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \geq 0$; These formulae determine the thermobehavior of the

$$\frac{L}{N K}$$



valid around system, for $\beta \rightarrow 0^+$ or $\beta \rightarrow 0^-$
here



- Exercise: ideal gas in constant $\vec{B}$ enclosed in a volume V
 find: $\uparrow$ average dipole moment $\uparrow$ Strong/Weak $\vec{B}$ fields
 $\uparrow$ magnetic susceptibility $\uparrow$ Large/small T , Curie law?

• ideal gas of indistinguishable spin- $\frac{1}{2}$ particles in $\vec{B}$

$$\ln \tilde{Z} = - \sum_n \ln(1 + e^{-\beta \epsilon_n} e^{\beta \mu}) = - \sum_n \ln(1 + e^{-\beta \epsilon_n(P) \pm \beta \tilde{\mu} B} e^{\beta \mu})$$

$$= \sum_n \ln(1 + e^{-\beta \epsilon_n(P)} e^{\beta \mu_+}) + \sum_n \ln(1 + e^{-\beta \epsilon_n(P)} e^{\beta \mu_-})$$

$$\mu_{\pm} = \mu \pm \tilde{\mu} B$$

$$z_{\pm} = e^{\beta \mu_{\pm}}$$

$$\ln \tilde{Z} = \sum_n \ln(1 + z_+ e^{-\beta \epsilon_n(P)}) + \sum_n \ln(1 + z_- e^{-\beta \epsilon_n(P)})$$

prev section

$$\Rightarrow J = -kT \ln \tilde{Z} \simeq (\text{deg}) \frac{kTV}{\lambda_T^3} (L_{i_{5/2}}(z_+) + L_{i_{5/2}}(z_-))$$

$$\lambda_T = \left(\frac{h^2}{2\pi m kT} \right)^{1/2}$$

$$\langle M \rangle = -\frac{1}{N} \left(\frac{\partial J}{\partial B} \right)_{T, \mu, V}, \quad \chi_T = \left(\frac{\partial \langle M \rangle}{\partial B} \right)$$

N_+ , N_- - Number that are parallel and antiparallel to $\vec{B}$

$$\langle M \rangle = \frac{\tilde{\mu}}{N} (N_+ - N_-)$$

$$\text{for small temps: } \lambda_T = \frac{\partial \langle M \rangle}{\partial B} = -\frac{V}{N} \frac{\tilde{\mu}}{\lambda_T^3} (\beta \tilde{\mu}) [L_{i_{5/2}}(z_+) + L_{i_{5/2}}(z_-)]$$

$$\simeq \frac{-2\beta \tilde{\mu}^2 V}{\lambda_T^3} L_{i_{5/2}}(-z) \Rightarrow \text{show Curie Law is valid.}$$